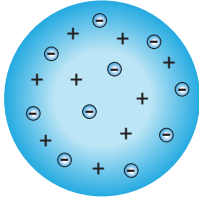
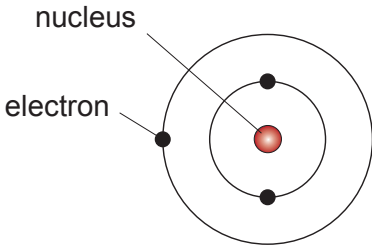
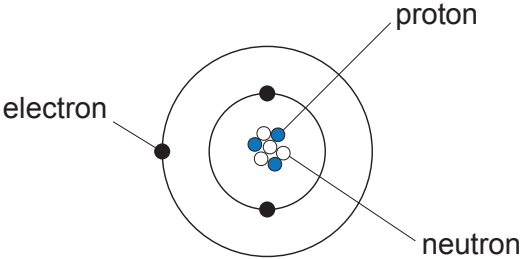


5. (a) For over two centuries, scientists have created different models of the atom. Our understanding of the atom has developed hugely over that time.

In the early 1800s, scientists already knew that:

- atoms are invisible
- atoms of one element are all the same
- atoms of different elements are different
- atoms combine to form compounds

The table shows some of the ideas that led to what we now know.

|                                                                                                                                                                                                     |                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p><b>J.J. Thomson (1897)</b></p> <ul style="list-style-type: none"> <li>• Atoms are made of positive matter</li> <li>• Negative particles are spread out throughout the positive matter</li> </ul> |    |
| <p><b>Ernest Rutherford (1911)</b></p> <ul style="list-style-type: none"> <li>• Protons are in a nucleus in the centre of the atom</li> <li>• Atoms are mostly empty space</li> </ul>               |  |
| <p><b>James Chadwick (1932)</b></p> <ul style="list-style-type: none"> <li>• Atoms have positive and negative particles, and they also have particles with no charge</li> </ul>                     |  |



- (i) Tick (✓) the statement which describes a **difference** between J.J. Thomson's model and Rutherford's model. [1]

neither model has any neutrons

☐

Thomson had electron shells in his model

☐

Thomson did not think atoms are mostly empty space

☐

- (ii) Tick (✓) the statement that does **not** describe Chadwick's model of the atom. [1]

electrons are in shells outside the nucleus

☐

atoms have equal numbers of protons and electrons

☐

atoms have equal numbers of protons and neutrons

☐

- (iii) Tick (✓) the **three** statements which correctly describe how our knowledge about atoms today is different to J.J. Thomson's model. [2]

electrons are outside the nucleus

☐

electrons are inside the nucleus

☐

there are more protons than electrons in an atom

☐

protons are in a nucleus in the centre of an atom

☐

atoms have particles with no charge

☐

(b) An atom of aluminium can be shown as  ${}_{13}^{27}\text{Al}$ .

State the number of protons and neutrons in this atom.

[2]

Number of protons .....

Number of neutrons .....

Examiner  
only

|   |
|---|
|   |
| 6 |

